

PyGameMaker — Tutorial 15: Fruit Fusion

— Worksheet

Name: _____ Date: _____ Do this worksheet after you have won and lost... actually, you cannot lose! Do it after you have won the game at least once.

Part A: Words

Match each word with what it means. Write the letter next to the number.

- 1 Fusion ____
 - 2 Held fruit ____
 - 3 If / Else ____
 - 4 Spawner ____
- **A.** A block that does one thing when something is true, and a different thing when it is not
 - **B.** An invisible object whose only job is to create other objects on a timer
 - **C.** Catching a matching fruit so it grows into the next, bigger fruit
 - **D.** The one fruit the basket is currently carrying, remembered in held_level

Part B: Which object and event?

Write the object and the event for each rule.

- 1 Turn into a strawberry when a matching cherry is caught: _____
- 2 Create a new cherry every 90 steps: _____
- 3 Remove a fruit that falls past the basket without being caught: _____
- 4 Show the YOU WIN! message: _____
- 5 Restart the game when SPACE is pressed on the win screen: _____

Part C: Think about it

- 1 held_level starts at 1 (a cherry). Why does catching a cherry while held_level is already 3 (an orange) only give 1 point instead of fusing?

- 2 What would happen if the orange collision's *Go to room room_win* action were accidentally placed in the *Otherwise* branch instead of the *If* branch?

- 3 Why does each fruit type need its own spawner object, instead of one spawner picking a random fruit type?

Part D: Try it

- ☐ The basket moves and stops
- ☐ Catching a matching cherry turns the basket into a strawberry and adds 10 points
- ☐ Catching a matching strawberry turns the basket into an orange and adds 20 points
- ☐ Catching a matching orange turns the basket into a watermelon, adds 50 points, and shows YOU WIN!
- ☐ A mismatched catch only adds 1 point and changes nothing else

Part E: Look back

- 1 What I am proudest of: _____
- 2 A bug I found and fixed: _____
- 3 What I would add next: _____